

Pranav Pappu

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Education

National University of Singapore

2025 – 2029

Bachelor of Computing (Honours) in Computer Science

AIE Scholarship – Selected as 1 of 20 recipients

Technical Skills

Languages/Runtime: Python • Zig • C++ • Rust • TypeScript/JavaScript • Node.js

Infrastructure/DevOps: Docker/Compose • GitHub Actions CI/CD • Fly.io • Linux

Backend/Data: FastAPI • PostgreSQL/pgvector • SQLAlchemy/Alembic • Pydantic

AI/Testing: PyTorch • Transformers • OpenAI API • MCP (stdio/HTTP) • pytest • Playwright

Experience

Founder, Tweakler

2026 – Present

- Founded Tweakler, a reference-led website generation system that helps agents avoid generic UI: from a one-line brief, it derives an audience's values and jobs-to-be-done, surfaces diverse permissioned design references, and transforms a selected reference into a distinct, editable site with brand-specific copy, assets, and visual direction. Piloted with studios including [Brightsea Studio](#).
- Built a [54-tool MCP workflow](#) and deterministic browser capture-to-code engine for rights-gated cloning, responsive site generation, transactional copy and asset replacement with rollback, and fail-closed desktop/mobile visual, accessibility, build, and type-check validation.

Research Assistant, National University of Singapore

Jan 2026 – Present

- Built a dynamic 3DGS benchmark with [11 YAML pipelines](#) across [6 method repos](#) and [3 datasets](#), plus a [6-metric evaluator](#) for repeatable quality/performance reports.

CCSGP Fellow (AssessMate), NUS Centre for Computing for Social Good

2025 – Present

- Building AssessMate, an AI case-notes copilot for structured documentation; piloted with [2 assessors](#) across multiple sessions to measure documentation-time reduction on PDPA-sensitive child-development data.

C++ and Python Specialist (Freelance), Outlier.ai – Remote

Dec 2024 – May 2025

- Evaluated C++/Python model outputs for correctness, edge-case coverage, and reasoning quality, contributing feedback for code-generation model improvement.

Projects

EvoServe – Harness Evaluation & Quality Control

2026 – Present

- Built EvoServe, a harness-quality and evaluation control plane for coding-agent systems: compares policy candidates, ingests captured runs, independently verifies evidence, and produces auditable reports for quality and regression review.
- Implemented a [10-command](#) validate/replay/score/verify/report CLI with [36 JSON schemas](#) and [24 Zig tests](#); fails closed on malformed, missing, ambiguous, mutated, or untrusted evidence while preserving explicit review before policy activation.

GPT-2 (124M) Recreation from Scratch

2024

- Recreated GPT-2 124M from scratch (tokenizer, transformer blocks, optimizer, mixed precision, and FineWeb Edu evaluation); trained on [10B tokens](#).

WikiBase – Source-Grounded Knowledge Platform

2026

- Built WikiBase, a student knowledge platform that turns PDFs and notes into cited Markdown wikis, grounded Q&A, and flashcards; architected its [FastAPI/PostgreSQL/pgvector](#) backend, idempotent [5-stage pipeline](#), Docker/Fly.io release infrastructure, and [GitHub Actions CI/CD](#) plus a fail-closed [487-test harness](#) covering migrations, security scans, and smoke checks.

Competitions

Replit Agent Hackathon

2026

- [1st place](#): Built Remember, an AR memory assistant with [4 AI workflows](#) for face recognition, speech transcription, LLM memory extraction, and voice Q&A over [128-float face embeddings](#) to a persistent personal knowledge graph for context-aware recall across repeated encounters.

Google DeepMind × Cactus (YC S25) Global Hackathon

2026

- [2nd place](#): Built SecureClaw, a privacy router for [12 PII types](#) with [3 routing modes](#), on-device/cloud inference, and a [30-case benchmark](#).